

Zhecheng Yu

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Experiences

- **Visiting Scholar** – *National University of Singapore, Singapore* Aug 2025 – Apr 2026 (expected)
- **BEng** – *Southeast University, Nanjing, China* Aug 2022 – Jun 2026 (expected)
- Training Base for Top Students in Computer Science (top 20 CS undergraduates) Avg. Score: 89.29

Research Interests

Computer Vision, Multimodal learning, Explainable AI

Publications

- [1] **Learning from Human Gaze: Human-like Robot Social Navigation in Dense Crowds** AAAI 2026
Zhecheng Yu, Yan Lyu, Chen Yang, Tao Chen, Yishuang Zhang, Bo Ling, Peng Wang, Guanyu Gao, Weiwei Wu, Brian Y. Lim

Projects

Imaginative XAI
Supervisor: Prof. Brian Y. Lim, NUS Aug 2025 – present

- Impute hard-to-acquire modalities via generation techniques to improve downstream decision-making performance.
- Synthesize intuitive non-verbal rationalizations (e.g., visual simulation) to lower the cognitive load of interpretation.
- Generate diverse counterfactuals to challenge user prior belief and reduce confirmation bias in human reasoning.

Gaze-Augmented AI
Supervisor: Prof. Yan Lyu, SEU & Prof. Brian Y. Lim, NUS Jun 2024 – present

- **Comprehensive Egocentric Gaze-Motion Dataset for Crowd Navigation Analysis** Nov 2025 – present
 - Develop a gaze-semantic system integrating gaze data with SOTA CV models to bridge the gap from pixels to semantics.
 - Integrate IMU data to capture head orientation and rotational dynamics, enabling a holistic analysis of visual exploration.
 - Conduct a comprehensive behavioral analysis across diverse demographics, navigational intents, and crowd densities.
- **From Gaze to Human-like Robot Crowd Navigation** Nov 2024 – Jul 2025
 - Collected an egocentric dataset featuring synchronized human gaze, video, and trajectory data in real-world crowds.
 - Developed a modular framework that predicts human-like gaze to identify socially saliency and guide navigation.
 - Demonstrated that incorporating human gaze improves both navigation performance and human alignment.
- **VR-Based Gaze Projection onto RGB-D Videos** Jun 2024 – Nov 2024
 - Developed eye tracking and gaze interaction functionalities in a virtual environment with a MR headset and Unity.
 - Integrate the MR headset with a depth camera into a multimodal data collection tool for real-world eye tracking.
 - Matched gaze vectors to 3D pixels by maximizing directional similarity, mapping 2D gaze points onto the RGB video.

Extracurricular Experiences

Head of the Organizational Department The CSE Students' Union, Southeast University Sep 2023 – Jun 2024

- Planned, organized and hosted large-scale campus events
- Demonstrated strong leadership and teamwork skills

Awards

- **Outstanding Research-based Final Year Project (FYP)** (top 24 undergraduates university-wide) Dec 2025
- **China-Singapore SIP International Scholarship** 10,000 CNY (1/304) Oct 2025
- **National Grand Prize** 2024 National English Competition for College Students (top 1‰) May 2024